The Multilateral Ledger: Civilizations as Networks of Reciprocal Relations

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15 October 2026

Abstract

The companion paper *The Essence of Civilization* develops a formal theory of civilization from the bilateral relation: two parties, one ledger, one set of thresholds. It states explicitly that the multilateral case is deferred. The present paper takes up that case. The extension is not cosmetic. Once relations exist among many parties, entirely new phenomena appear that have no bilateral counterpart: chains of layered extraction, races to exploit a shared host, the aggregation problem of collective voice, the emergence of the center as an endogenous actor, legitimacy as a public good, and the cascade collapse of networks. The paper derives each of these formally. Its central results are four. First, in an m-party relation with Cobb–Douglas technology, the voluntary contribution equilibrium exists and is unique if and only if the sum of the members' output elasticities falls below unity; when it does not, production must be organized by command, and when it does, per-member surplus under equal shares decays as $m^{-1/D}$, where D is the returns-to-scale deficit—a precise law of imperial overstretch. Second, m predators racing on a common host behave as a single predator with discount rate multiplied by m: the unique open-loop Nash steady state is $K^* = (1 - mr)/(m + 1)$, so the host is destroyed whenever $mr \geq 1$; patient predators conserve only in the absence of competitors, and each predator's equilibrium payoff falls in m, so the race is Pareto-inferior for the predators themselves—the ledger basis of the predatory cartel we call a state. Third, collective voice saturates in group size under coordination friction, so extraction is lowest for small organized groups and highest for large unorganized classes; communication technology is therefore a justice technology, and its suppression a legitimacy technology. Fourth, any relation that moves the thresholds of an actor who is not a party to it creates *unledgered* injustice; legitimate multilateral order requires that standing match the footprint of a relation's effects—the formal content of "no taxation without representation," and its extension to externalities and future generations. The paper closes with network dynamics in which collapse becomes a cascade, and shows that the systemic risk of predation scales with the centrality of the host.

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1 Introduction

The bilateral theory of *The Essence of Civilization* rests on the relation $e = (i, j)$: two parties, contributions, output, shares, an exit cost, a voice, two thresholds. From that smallest unit it derives a classification of relations, an extraction ceiling, a conservation principle for predation, a legitimacy constraint, and a collapse theorem. Throughout, the paper notes that "relations may be bilateral or, by extension, multilateral; the bilateral case is treated here for tractability." This paper removes that restriction. The restriction matters. A theory that stops at pairs cannot see the phenomena that dominate the history of large orders: the chain of intermediaries through which tribute travels from province to capital; the race by which many extractors jointly consume a common pasture, fishery, or atmosphere; the impossibility of getting a peasantry to act in concert; the rise of the center—the palace, the temple, the state—as a party that did not exist in the bilateral account and yet comes to dominate every ledger it touches; the free-riding on legitimacy that forces orders to substitute surveillance for belief; the cascade by which the fall of one node drags down its neighbors until the whole network is consumed. Each of these is irreducibly multilateral. None can be written down with two actors without smuggling in a third. The paper proceeds in one movement: it rebuilds the primitives for n parties, then re-derives, in order, the classification (section 3), the viability set (section 4), the distribution problem (section 5), the sustainability of cooperation (section 6), the aggregation of voice (section 7), the competition among extractors (section 8), the emergence of the center (section 9), the public-good structure of legitimacy (section 10), the multilateral theory of justice (section 11), alliances and war (section 12), and cascade dynamics (section 13). A summary of results is given in the Appendix. Four results deserve mention in advance.

1. **The overstretch law.** In an m-party relation with Cobb–Douglas technology and returns-to-scale deficit $D = 1 - \sum_i \alpha_i > 0$, equal shares deliver per-member net gain proportional to $m^{-1/D}$. A civilization can absorb members only while this decay keeps every member above its participation threshold; this yields a maximum sustainable coalition size (theorem 4.7).

2. **The predation race.** m predators extracting a common logistic host in open-loop Nash equilibrium converge to the unique steady state $K^*(m, r) = (1 - mr)/(m + 1)$. The host survives iff $r < 1/m$ (theorem 8.1). Competition acts as an additive premium on impatience. Moreover each predator's equilibrium payoff is strictly decreasing in m: the race is Pareto-inferior for the racers, which is why extractors historically converge on a monopolist (proposition 8.5).

3. **The coordination ceiling.** Collective voice of a class of size m is $V(m) = mv/(1 + \gamma(m - 1))$, where γ is the coordination friction. For $\gamma \geq 1$, V is nonincreasing in m: large unorganized classes face strictly higher extraction than small organized ones with identical individual endowments (theorem 7.3).

4. **The standing principle.** A relation that moves the thresholds of an actor who is not a party to it creates injustice that no bilateral ledger records. Multilateral legitimacy requires that standing match the footprint of effects (definition 5.8, proposition 5.9).

2 Multilateral Primitives

2.1 Actors, Relations, and the Graph

Let $N = \{1, \ldots, n\}$ be a finite set of actors. A *relation* is now a subset $e \subseteq N$ with $|e| \geq 2$; its *arity* is $|e|$. Write E for the set of relations and $\mathcal{G} = (N, E)$ for the resulting *civilizational graph*. Actors may participate in many relations; an actor's *degree* $d_i = |\{e : i \in e\}|$ counts its participation. The bilateral theory is the restriction of the present one to relations of arity two. Each relation e is characterized by:

- **Contributions** $c_i^e \in [0, \bar{c}]$ for each $i \in e$, in units of reproductive capacity.
- **Output** $Q_e = \eta_e f_e\big((c_i^e)_{i \in e}\big)$, with $\eta_e > 0$.
- **Shares** $s_i^e \geq 0$, $\sum_{i \in e} s_i^e \leq 1$; the residual is friction or third-party capture.
- **Exit cost** $k_i^e \geq 0$ and **voice** $v_i^e \geq 0$.

Convention 2.1 (Exit cost). The exit cost k_i is the loss incurred upon exit: the value of the best outside option, net of the cost of reaching it. Actor i therefore prefers continuation to exit whenever its net gain exceeds $\bar{a}_i - k_i$. This convention makes the comparative statics of (14) match those of the bilateral paper's prose: higher exit costs raise the extraction ceiling.

2.2 Net Gain and the Aggregate Ledger

Definition 2.2 (Net gain)**.** The net gain to actor i in relation $e \ni i$ is

$$n_i^e \;=\; s_i^e\, Q_e \;-\; c_i^e. \tag{1}$$

Actor i's *position* in the ledger is the vector $\mathbf{n}_i = (n_i^e)_{e \ni i}$, together with its aggregate $n_i = \sum_{e \ni i} n_i^e$.

The two thresholds of the bilateral theory carry over unchanged: the participation threshold $\bar{a}_i$ and the reproduction threshold ρ_i, with capacity dynamics

$$\dot{K}_i \;=\; g_i K_i\Big(1 - \tfrac{K_i}{K_i^{\max}}\Big) \;+\; n_i \;-\; \zeta_i K_i \;-\; \phi_i C_i. \tag{2}$$

What is new is the *coupling*: n_i now aggregates over many relations, and the failure of any relation moves every participant's capacity. The bilateral theory had one coupling; the multilateral theory has a network of them.

2.3 Coalitions

Definition 2.3 (Coalition and characteristic function)**.** A *coalition* is a subset $S \subseteq N$. Let

$$v(S) \;=\; \max_{(c^e, s^e)_{e \subseteq S}} \; \sum_{i \in S} \; \sum_{e \ni i,\, e \subseteq S} n_i^e \tag{3}$$

be the maximum joint net gain S can achieve using only relations internal to itself. We call v the *characteristic function* of the order.

Remark 2.4. v is defined over internal relations only. Relations that cross the boundary of S—between $i \in S$ and $j \notin S$—are precisely the externalities that the bilateral theory cannot see and that Section 5 will make central.

3 Classification: Pairs, Chains, and Systems

3.1 Relation-Level Classification

The bilateral definitions of symbiosis, parasitism, and predation apply relation by relation, now over $|e|$ parties.

Definition 3.1 (Multilateral classification)**.** Relation e is *symbiotic* iff $n_i^e \geq \bar{a}_i$ for all $i \in e$; *parasitic* iff no party is below its reproduction threshold ($n_i^e \geq \rho_i$ for all i) and some party satisfies $\rho_j \leq n_j^e < \bar{a}_j$; *predatory* iff some party satisfies $n_j^e < \rho_j$.

The civilizational classification aggregates conservatively: the order inherits the worst category present, exactly as in the algorithmic paper.

3.2 Chains

The genuinely new object is the *chain*: a sequence of relations $e_1, \dots, e_L$ through which value is extracted in layers—province to regional governor to capital; tenant to landlord to crown; user to platform to advertiser; borrower to bank to bondholder.

Definition 3.2 (Extraction chain)**.** An *extraction chain* of length L is a sequence of parties $0, 1, \dots, L$ and relations $e_\ell = (\ell - 1, \ell)$, $\ell = 1, \dots, L$, in which party ℓ extracts from party $\ell - 1$ at rate τ_ℓ and forwards a fraction of what it collects, the forwarding rule being endogenous to the local ledger of relation e_ℓ.

Proposition 3.3 (Layered extraction)**.** *Let the bottom party* 0 *produce output* Q *and let each layer* ℓ *extract a rate* τ_ℓ *from the layer below. The fraction of output reaching the top is* $\prod_{\ell=1}^{L} \tau_\ell$*, and the bottom party retains* $(1 - \tau_1)Q$ *before its own contribution cost. Holding the extraction reaching the top fixed, a longer chain strictly reduces the bottom party's net gain.*

Proof. The extraction fractions multiply along the chain. With the top extraction $X = \prod_\ell \tau_\ell < 1$ fixed, the symmetric layer rate $X^{1/L}$ rises toward one as L grows, so the bottom's retained fraction $1 - \tau_1$ falls. □

Proposition 3.4 (The weakest link)**.** *Suppose each layer* ℓ *is locally feasible only while its own legitimacy constraint holds,* $\Phi(G_\ell, L_\ell) \le \tau_\ell Q_\ell$*. Then the chain fails at the first layer whose local constraint binds: that layer ceases to forward, every layer above it is starved of inflow, and every layer below it is no longer extracted from. The location of failure is set by local constraints and is independent of distance from the top or the bottom.*

Proof. Each layer's constraint is local: it involves that layer's injustice gap and legitimacy, not those of other layers. A binding layer stops forwarding, so all layers above it lose their input and the chain terminates there, wherever in the hierarchy that layer sits. □

Corollary 3.5 (Peripheral revolution)**.** *Chains of extraction are broken from the weakest link, which is typically not the top. The historical pattern in which revolts begin at the periphery of empires—Judaea, Gaul, the Andes—rather than in the capital, is the signature of Proposition 3.4.*

3.3 Systemic Classification

Definition 3.6 (Systemic severity)**.** Let π_ρ be the fraction of actors with $n_i < \rho_i$ and $\pi_{\bar{a}}$ the fraction with $\rho_i \le n_i < \bar{a}_i$. The order is *systemically predatory* if π_ρ exceeds a threshold or if predation concentrates on actors whose removal fragments $\mathcal{G}$; *systemically parasitic* if $\pi_{\bar{a}}$ is large while π_ρ is small.

Remark 3.7. Centrality matters as much as prevalence. Predation on a peripheral actor is a local fact; predation on a hub is a systemic one, because the hub's capacity sustains every relation it touches. Section 13 makes this precise.

4 The Viability Set in Many Dimensions

4.1 The Cobb–Douglas Relation

Let e have parties $N_e = \{1, \dots, m\}$ (arity m to avoid clashing with the order's size n), technology

$$Q \;=\; \eta \prod_{i=1}^{m} c_i^{\alpha_i}, \qquad \alpha_i > 0, \qquad D \;=\; 1 - \sum_{i=1}^{m} \alpha_i. \tag{4}$$

Define the matrix $M \in \mathbb{R}^{m \times m}$ by $M_{ii} = 1 - \alpha_i$ and $M_{ij} = -\alpha_j$ for $i \neq j$.

Lemma 4.1 (Existence and uniqueness)**.** *The one-shot relation game has a unique interior Nash equilibrium of contributions if and only if $D > 0$. In that case, writing $x_i = \log c_i^*$ and $b_i = \log(s_i \alpha_i \eta)$, the equilibrium solves*

$$Mx \;=\; b, \qquad \textit{hence} \qquad c_i^* \;=\; \prod_{j=1}^{m} (s_j \alpha_j \eta)^{(M^{-1})_{ij}}, \tag{5}$$

with $M^{-1} \geq 0$ elementwise and equilibrium net gain

$$u_i^* \;=\; c_i^* \, \frac{1 - \alpha_i}{\alpha_i}. \tag{6}$$

Proof. The first-order condition for player i is $s_i \alpha_i \eta \prod_j c_j^{\alpha_j} = c_i$, which in logs reads $x_i - \sum_j \alpha_j x_j = b_i$, i.e. $Mx = b$. Every row sum of M equals D, so M is strictly row-diagonally dominant precisely when $D > 0$; it is then a nonsingular M-matrix with nonnegative inverse, giving (5). Substituting the first-order condition into $u_i = s_i Q - c_i$ gives $u_i^* = c_i^*(1/\alpha_i - 1)$. □

Definition 4.2 (Viability set)**.** For a relation of arity m with share vector $s \in \Delta^{m-1}$, the *viability set* is

$$\mathcal{V} \;=\; \big\{\, s \in \Delta^{m-1} :\; u_i^*(s) \geq \bar{a}_i \text{ for all } i \in N_e \,\big\}, \tag{7}$$

the shares under which every party prefers participation to its outside option.

Remark 4.3. Since $M^{-1} \geq 0$ elementwise, each u_i^* is nondecreasing in every share s_j; the viability set is therefore an upper set in the share lattice. Enlarging any share can only add allocations to $\mathcal{V}$, never remove them.

Proposition 4.4 (Command threshold)**.** *If $\sum_i \alpha_i \geq 1$, no interior contribution equilibrium exists: returns to coalition size are too strong for voluntary contribution to stabilize. Production at that arity must be organized by imposed contribution floors—the formal domain of corvée, slavery, conscription, and command hierarchy.*

Remark 4.5. This is the multilateral face of the bilateral result that parasitism requires constraint. The bilateral theory found that free choice cannot generate parasitism. The multilateral theory adds that free choice cannot even generate *production* once the coalition's combined output elasticities exceed unity. Hierarchy is not only a technology of extraction; it is a technology of scale.

4.2 The Overstretch Law

Lemma 4.6 (Homogeneity)**.** *The equilibrium contribution vector c^* is homogeneous of degree $1/D$ in the share vector s: $c^*(\lambda s) = \lambda^{1/D} c^*(s)$. Since $M\mathbf{1} = D\mathbf{1}$, the row sums of M^{-1} all equal $1/D$.*

Proof. Scaling s by λ shifts b by $\log\lambda \cdot \mathbf{1}$, so x shifts by $\log\lambda \cdot M^{-1}\mathbf{1}$. The row-sum claim follows from $M\mathbf{1} = D\mathbf{1}$. □

Theorem 4.7 (Overstretch)**.** *Under equal shares $s_i = 1/m$ in the symmetric case $\alpha_i = \alpha$, $\eta_i = \eta$, with $D = 1 - m\alpha$,*

$$u^*(m) \;=\; \frac{1 - \alpha}{\alpha} \left(\frac{\alpha \eta}{m} \right)^{1/D}. \tag{8}$$

Then:

(i) In the dilute regime $m \geq \alpha\eta$, u^* is strictly decreasing in m, since $\frac{d}{dm} \log u^* = \big[\alpha \log(\alpha\eta/m) - D/m\big]/D^2 < 0$.

(ii) Viability at arity m—every member above its participation threshold $\bar{a}$—holds iff $u^*(m) \geq \bar{a}$. Whenever the relation is viable at some $m_0 \geq \alpha\eta$, there is a finite maximum sustainable size $m^* = \max\{m : u^*(m) \geq \bar{a}\}$, and m^* is increasing in productivity η and decreasing in $\bar{a}$.

(iii) Equivalently, the viability condition $u^*(m) \geq \bar{a}$ is a lower bound on the returns-to-scale deficit: the relation is viable iff $D \geq \underline{D}(m)$ for the threshold $\underline{D}(m)$ solving $u^* = \bar{a}$. A technology with too thin a deficit cannot sustain the coalition under equal shares.

Proof. The formula is Lemma 4.6 evaluated at $s_i = 1/m$ with the symmetric inverse $M^{-1} = I + \frac{\alpha}{D}J$. Claim (i) follows by differentiation; claim (ii) is monotonicity plus the intermediate value theorem; claim (iii) restates (ii) using the strict increase of u^* in D on the dilute regime. □

Corollary 4.8 (Technology sets the scale of politics)**.** *The maximum sustainable coalition size m^* is increasing in productivity η and decreasing in the participation threshold $\bar{a}$, and shrinks as the returns-to-scale deficit thins. Orders built on technologies with thin deficits cannot sustain large voluntary coalitions; they must either raise productivity, lower members' outside options, or shrink.*

Illustration 4.9. The classical pattern of imperial overstretch—Assyria, Rome, the Habsburgs, the British Empire—is the crossing of Theorem 4.7: the frontier of membership moves outward until the per-member return to membership falls below the participation threshold, at which point secession becomes jointly rational and the center faces simultaneous peripheral exits.

5 Distributing the Surplus: Core, Shapley, and Standing

5.1 From Bilateral Fairness to the Core

The bilateral theory defines the fair share by maximizing the joint gain of the pair. With n parties no pairwise criterion suffices, because any allocation that pleases one coalition may be blocked by another.

Definition 5.1 (Core)**.** The *core* of the order is the set of allocations $u = (u_i)$ satisfying

$$\sum_{i \in S} u_i \;\geq\; v(S) \quad \text{for all } S \subseteq N, \qquad \sum_{i \in N} u_i \;=\; v(N). \tag{9}$$

An allocation in the core admits no blocking coalition.

Definition 5.2 (Shapley value)**.** The *Shapley value* $\varphi(v)$ assigns to each actor i its average marginal contribution over all orders of coalition formation:

$$\varphi_i(v) \;=\; \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(n - |S| - 1)!}{n!} \Big[v(S \cup \{i\}) - v(S) \Big]. \tag{10}$$

Theorem 5.3 (Multilateral fair share)**.** *If v is supermodular (coalition members are complements, so marginal contributions rise with coalition size), then $\varphi(v)$ belongs to the core. In the symmetric case $\varphi_i = v(N)/n$ for all i: the multilateral form of equal division, to which the bilateral fair-share rule reduces when $m = 2$ and the two parties are symmetric.*

Proof. Supermodularity of v implies convexity, and the Shapley value of a convex game lies in the core [12]; symmetry is immediate from the definition. □

Definition 5.4 (Multilateral injustice gap)**.** The injustice gap of actor i is $G_i = \max\big(0, \varphi_i - u_i\big)$, and aggregate injustice is $G = \sum_i G_i$.

5.2 Core Emptiness and the Impossibility of Reform

Proposition 5.5 (Core emptiness)**.** *Cross-boundary effects generate partition-function externalities, so the game v among coalitions need not be supermodular or balanced. When it is not, the core may be empty: no allocation is immune to blocking.*

Proof. A coalitional game has a nonempty core if and only if it is balanced; when relations internal to S change the payoffs available to coalitions outside S, v is a general partition function, which is not balanced in general. Emptiness is therefore possible once externalities cross coalition boundaries. □

Remark 5.6. An empty core is a precise formal object with a historical reading: there is no stable division of the surplus—every allocation will be overthrown by some coalition. The order must then do one of three things: empower an enforcer who makes side payments credible (the center of section 9); enlarge the set of parties until externalities are internalized (the standing principle below); or fragment until the residual coalitions each have nonempty cores.

5.3 The Principle of Standing

The deepest multilateral correction to the bilateral theory is not about numbers of parties; it is about *who is missing.*

Definition 5.7 (Affected set)**.** The *affected set* of relation e is

$$A(e) \;=\; \big\{\, i \in N :\; e \text{ moves } \bar{a}_i \text{ or } \rho_i \text{ or } K_i \,\big\}. \tag{11}$$

Definition 5.8 (Standing)**.** Relation e satisfies *standing* iff $A(e) \subseteq e$: every actor whose thresholds the relation moves is a party to it, with share-setting voice in its terms.

Proposition 5.9 (Unledgered injustice)**.** *If $i \in A(e) \setminus e$, then i's injustice gap G_i increases without appearing in any ledger that the bilateral accounting records. No pairwise audit detects the transfer; only the multilateral affected-set test does. An order in which standing systematically fails is parasitic at the systemic level even if every recorded relation is pairwise symbiotic.*

Proof. i bears a transfer through e's externality, so n_i falls by the externality's magnitude; since i is not a party, $s_i^e = 0$ and the transfer never enters (1) for any recorded relation. The gap G_i therefore accrues invisibly. □

Corollary 5.10 (The generalization of representation)**.** *"No taxation without representation" is the standing principle applied to the relation of taxation: the taxed are affected, hence must be parties. The principle extends without strain to every externality: pollution moves the thresholds of downwind non-parties; carbon moves the thresholds of future generations who cannot be parties at all; unpaid care labor moves the thresholds of caregivers absent from the product accounts. Each is a violation of standing, hence a structural source of injustice that the record does not contain and that ordinary auditing cannot find.*

Remark 5.11. The standing principle supplies the multilateral form of the hidden-history program of the algorithmic paper: search not only for gaps between monuments and recorded labor, but for threshold-moving non-members—actors the ledger omits while the physics of the relations does not.

6 Cooperation at Scale

6.1 Deviations by Coalitions

The bilateral patience threshold came from a two-player grim trigger. With n players the relevant deviation is not individual but *coalitional*: any subset S may jointly deviate while the rest punishes.

Definition 6.1 (Multilateral patience threshold)**.** Let u_i^C be the cooperative payoff, $u_i^{D(S)}$ actor i's payoff when coalition S jointly deviates, and u_i^P the punishment-phase payoff. Define

$$\delta^*(n) \;=\; \max_{\emptyset \neq S \subseteq N} \; \max_{i \in S} \frac{u_i^{D(S)} - u_i^C}{u_i^{D(S)} - u_i^P}. \tag{12}$$

Proposition 6.2 (Coalitional sustainability)**.** *The cooperative profile is subgame-perfect in the repeated multilateral game if and only if $\delta \geq \delta^*(n)$.*

Proof. Grim-trigger logic applied to each coalition's deviation inequality, exactly as in the bilateral case, then maximized over S. □

Remark 6.3. The critical deviations are typically small coalitions—pairs and triples—because the joint gain to a large coalition must be divided among many members; only the members' internal split, not the coalition's total gain, determines the temptation to deviate.

6.2 Monitoring and the Ledger Technology

The bilateral theory treated monitoring as a parameter. At scale it becomes an institution.

Assumption 6.4 (Partial monitoring)**.** Each relation's contributions are observed with probability $q \in (0, 1]$; undetected deviations occur with probability $1 - q$.

Proposition 6.5 (Scale raises the patience requirement)**.** *Under partial monitoring, $\delta^*(n)$ is nondecreasing in n: enlarging the order adds potential deviating coalitions without removing any, so sustained cooperation requires at least as much patience.*

Proof. Enlarging N enlarges the set of coalitions in (12); each coalition's deviation ratio is unchanged, so the maximum over a larger set cannot fall. □

Remark 6.6. The monotonicity is weak because a newly available coalition may deviate less profitably than the old maximum. Normally some added deviation is strictly more profitable, so δ^* rises; holding it fixed then requires monitoring quality q to rise with n, which is precisely Corollary 6.7.

Corollary 6.7 (Accounting as civilizational technology)**.** *An institution that raises q—census, cadastral survey, temple ledger, bureaucratic record, audit—raises the sustainable size of the cooperative order. Such an institution is worth its administrative cost M precisely when the surplus it unlocks—through the cooperation it makes sustainable—exceeds M. The theory therefore predicts that writing and accounting appear first as technologies of* scale*, not of expression: the earliest cuneiform is a ledger.*

Illustration 6.8. Uruk's temple accounts, Rome's census and annona registers, and the modern tax authority share one function in this framework: they raise q so that $\delta^*(n)$ falls enough for the order of that size to remain cooperative.

7 Voice, Coordination, and the Class Ceiling

7.1 The Aggregation Problem

In the bilateral theory, voice v_i was an attribute of an individual. In a multilateral order, the actors who most need voice—the many, the taxed, the unorganized—can exercise it only collectively, and collective action is itself a public good.

Definition 7.1 (Coordination friction)**.** Let a class of size m have identical members with individual voice v. Exercising collective voice requires organization, which suffers free-riding. Let $\gamma > 0$ denote the *coordination friction*: the per-member cost premium of acting together. Effective class voice is

$$V(m) \;=\; \frac{m\,v}{1+\gamma(m-1)}. \tag{13}$$

Lemma 7.2. *V is strictly increasing in m iff $\gamma < 1$, constant iff $\gamma = 1$, and strictly decreasing in m iff $\gamma > 1$. In all cases $V(m) \le \max(v,\, v/\gamma)$.*

Proof. $V'(m) = v(1-\gamma)/(1+\gamma(m-1))^2$; the bound follows by taking limits. □

7.2 The Extraction Ceiling for Classes

Suppose a principal faces a class of size m whose members are identical with exit cost k and effective voice $V(m)$. Two ceilings bind. The exit ceiling of Convention 2.1 is

$$\tau^{\text{exit}} \;=\; 1 - \sqrt{2\big(\bar{a}-k\big)_+}, \tag{14}$$

increasing in k: a captive class can be held to a higher rate. Separately, under generalized Nash bargaining the principal's equilibrium extraction is strictly decreasing in the class's effective voice $V(m)$, since voice raises the class's disagreement payoff. The sustainable rate is the lesser of the monopoly rate, (14), and the bargained rate.

Theorem 7.3 (Coordination ceiling)**.** *For $\gamma \ge 1$, extraction into the class is nondecreasing in class size, and strictly increasing wherever the voice constraint binds: V is nonincreasing in m by Lemma 7.2, and the bargained ceiling is decreasing in V. Holding individual endowments $(v, k, \bar{a})$ fixed, a large unorganized class is extracted from no less heavily than a small organized one, and strictly more heavily once $\gamma > 1$.*

Proof. By Lemma 7.2, V is nonincreasing in m for $\gamma \ge 1$; the bargained ceiling is decreasing in V, hence nondecreasing in m. The sustainable rate is the minimum of the two ceilings; since the exit ceiling is constant in m, the minimum inherits the monotonicity of the bargained ceiling, and rises strictly wherever that ceiling binds. □

Corollary 7.4 (Selective incentives as justice technology)**.** *Institutions that lower γ—unions, parties, associations, churches, cooperatives, and the printing press and telegraph and network that let them organize—raise V and lower extraction. Their suppression raises γ and is, in this vocabulary, an extraction policy.*

Corollary 7.5 (Divide and rule re-specified)**.** *Splitting one class of size m into r subclasses of size m/r while leaving within-class coordination intact does not lower total effective voice: $rV(m/r) = mv/\big(1+\gamma(m/r-1)\big) > V(m)$ for every $\gamma > 0$. Fragmentation of the class is therefore not itself an extraction policy. What pays the extractor is raising γ itself—by identity engineering, surveillance, and disinformation—or blocking organization within each subclass. The historical art of divide-and-rule is the deliberate engineering of coordination friction, not of class size.*

Illustration 7.6. Peasants, a class of enormous size and low γ-technology, were taxed to the ceiling across agrarian history; merchant guilds and later joint-stock companies, small and cheaply coordinated, extracted privileges instead of bearing them. The labor movements of the nineteenth century are legible in this framework as a single shock to γ, and their repression as the counter-shock.

Remark 7.7 (Managed decay, multilateral). The contemporary observation of the companion essays—legitimacy can decay without conflict—now has a mechanism. Conflict itself is a public good: every grievant benefits from a revolt none of them individually starts. When γ is high, V is low, and grievance does not translate into collective action. A communication technology that lowers γ is therefore a legitimacy event: the same G produces conflict that yesterday it did not, precisely because coordination became cheap.

8 Many Predators: Chains, Races, and Conservation Failure

8.1 The Race for the Host

The conservation principle of the bilateral theory states that a patient predator conserves its host: $K^*(r) = (1-r)/2$. The result assumed a sole predator. Most hosts in history have been shared: pastures, fisheries, provincial tax bases, slave catches, and the atmosphere. Let the host grow logistically, $\dot{K} = K(1-K) - \sum_{i=1}^{m} \tau_i K$, and let m predators choose extraction rates $\tau_i \geq 0$ to maximize $\int_0^\infty e^{-rt}\tau_i K\, dt$, each treating the others' rates as given (open-loop Nash).

Theorem 8.1 (Predation race). *The symmetric open-loop Nash equilibrium of the m-predator race has the unique steady state*

$$K^*(m,r) \;=\; \frac{1-mr}{m+1}, \tag{15}$$

with per-predator extraction $\tau^* = (1+r)/(m+1)$ and aggregate extraction $m\tau^* = 1-K^*$. The host is depleted, $K^* \leq 0$, whenever

$$mr \;\geq\; 1. \tag{16}$$

Proof. Predator i's current-value Hamiltonian, given rivals' aggregate extraction $\bar{\tau} = \sum_{j\neq i}\tau_j$, is $H_i = \tau_i K + \mu_i\big[K(1-K-\bar{\tau}-\tau_i)\big]$. The singular condition $\partial H_i/\partial\tau_i = K(1-\mu_i) = 0$ gives $\mu_i = 1$. The costate equation on the singular arc, $r = \tau_i + (1-2K-\bar{\tau}-\tau_i)$, reduces to

$$r \;=\; 1 - 2K - \bar{\tau}. \tag{17}$$

Symmetry gives $\bar{\tau} = (m-1)\tau$; stationarity of the host gives $\tau = (1-K)/m$. Substituting into (17) and solving yields (15). Feasibility $K^* \geq 0$ is exactly $mr \leq 1$. □

Corollary 8.2 (Strategic impatience). *Along* (17)*, rivals' extraction $\bar{\tau}$ enters as an additive premium on the predator's discount rate: each predator behaves as though its rate were $r_{\mathit{eff}} = r + \bar{\tau}$. Competition converts patience into impatience without any change in preferences.*

Corollary 8.3 (Critical coalition size). *For any discount rate $r > 0$ there is a critical number of predators $m_{\mathit{crit}} = \lfloor 1/r \rfloor$ beyond which the host is destroyed. A single very patient predator ($r \ll 1$) conserves; a sufficiently large group of such predators annihilates the same host.*

Corollary 8.4 (The commons, derived). *The tragedy of the commons is not a failure of patience but a consequence of numbers: it is the bilateral conservation principle with $m > 1$. Hardin's pasture [6] is* (15) *with r small and m large enough that $mr > 1$; Ostrom's communities [11] are the institutional creation of a center—or of rules that function as one—which converts the race into a single-predator problem.*

8.2 The Predators' Dilemma

Proposition 8.5 (Pareto-inferior race)**.** *Each predator's equilibrium payoff is*

$$u^*(m) \;=\; \tau^* K^* \;=\; \frac{(1+r)(1-mr)}{(m+1)^2}, \tag{18}$$

which is strictly decreasing in m for all $m \geq 1$ whenever $r < 1$. Every predator prefers the outcome with fewer predators; the race is Pareto-inferior for the racers themselves.

Proof. The derivative of $(1-mr)/(m+1)^2$ has the sign of $r(m-1)-2 < 0$ for $m \geq 1, r < 1$. □

Corollary 8.6 (Predatory cartel)**.** *Because every predator gains from the exclusion of rivals, extractors converge on monopolization: warlords on a king, mafias on territories, firms on regulation, empires on spheres of influence. The state is, among other things, a cartel agreement among predators, enforced by the very monopoly it establishes.*

Remark 8.7. This corollary and the protection-racket reading of the state are complementary: the host gains survival from monopolization (section 9), and so do the predators. Both pressures push the same direction, which is why centers arise so reliably and why their absence is so reliably followed by what the chronicles call dark ages.

8.3 Chains Revisited

Proposition 8.8 (Layered conservation failure)**.** *Suppose each of L chain layers is itself a race among m_ℓ predators. Then the host feeding layer ℓ is depleted whenever $m_\ell r_\ell \geq 1$, and the failure of any layer's host terminates the chain above it. Chain length multiplies the number of racing parties: an empire with L layers of provincial extraction faces L conservation constraints simultaneously, and its stability requires $m_\ell r_\ell < 1$ at every layer.*

9 The Emergence of the Center

9.1 The Center as Endogenous Actor

The bilateral theory had no center. The multilateral theory generates one: *the center* is an actor $C \in N$ that emerges to do what no pair can: monopolize predation, enforce transfers, supply monitoring, and provision public goods.

Definition 9.1 (Center)**.** A *center* is an actor that (i) extracts a fee $f \geq 0$ from every relation it governs; (ii) suppresses extraction races by imposing a single extraction rate; (iii) raises monitoring q; and (iv) raises legitimacy L of the order it governs.

Theorem 9.2 (Value of the center)**.** *The fee the center can sustainably charge is bounded by the value of what it provides:*

$$f \;\leq\; f^{\max} \;=\; \underbrace{\big[u^*(1,r) - u^*(m,r)\big]}_{\text{conservation of the race}} + \underbrace{\Delta_{coop}}_{\text{expanded cooperation}} + \underbrace{\Delta_{monitor}}_{\text{lower } \delta^*}, \tag{19}$$

where every term is a per-period flow, and separately by a legitimacy constraint $f \leq \bar{f}(L)$—the multilateral counterpart of the bilateral feasibility condition. The center is viable iff $f \leq \min\{f^{\max}, \bar{f}(L)\}$.

Proof. Direct aggregation of the three services identified in preceding sections: monopolization restores the single-predator conservation of section 8, monitoring raises q and thereby lowers $\delta^*(n)$ (proposition 6.5), and enforcement makes side-payments credible, enlarging the implementable set toward the core (section 5). No fee exceeding the sum of these services can be sustained, since the governed could do as well without the center. □

Proposition 9.3 (Comparative static of centrality)**.** *The conservation term in* (19) *is increasing in m: the more fragmented the extractors, the more valuable their monopolization. Strong centers are therefore endogenous to fragmentation.*

Proof. $u^*(1, r) - u^*(m, r) > 0$ and increasing in m by Proposition 8.5. □

Illustration 9.4. The pattern in which warlords welcome a unifier, and in which fragmented polities—post-Roman Europe, Warring States China, shogunal Japan—either centralize or are conquered by a neighbor that has, is the comparative static of Proposition 9.3 running across centuries.

9.2 The Guardian Problem

The center is itself an actor with its own discount rate.

Definition 9.5 (Horizon of the center)**.** Let the center's effective discount rate be $r_C = r + h$, where h is the hazard that it is overthrown, expropriated, or loses office.

Proposition 9.6 (Looting and infrastructure)**.** *A center with $r_C < 1$ conserves the host it governs: it builds, administers, invests. A center with $r_C \geq 1$ loots: it extracts to the destruction threshold and departs. Coups, succession crises, and revolutions raise h temporarily; the reconstruction that follows lowers it. The difference between a dynasty that builds canals and one that empties the treasury is not virtue but h.*

Proof. The center is a single predator after monopolization; apply the bilateral conservation principle with discount rate r_C. □

Corollary 9.7 (Institutions as horizon-lengtheners)**.** *Rotation, term limits, succession rules, legal personality, and credible commitment devices lower the center's effective h by binding its future selves. Institutions are the technology by which an order raises the patience of its own government.*

10 Legitimacy as a Public Good

10.1 Provision and Free-Riding

The bilateral theory treats legitimacy L as a state variable fed by ideological investment I. Multilaterally, I is a public good: one actor's belief does not diminish another's, while the cost of producing belief falls on producers.

Proposition 10.1 (Under-provision of legitimacy)**.** *Voluntary ideological investment is underprovided: each producer bears the full cost and captures a $1/n$ share of the legitimacy benefit. The equilibrium level of I falls as n rises, holding individual preferences fixed.*

Proof. Standard public-good free-riding: with linear provision, Nash investment is the sum of individually justified contributions, each scaled by the private share of the benefit. □

Corollary 10.2 (Coercion substitutes for belief)**.** *An order too large or too diverse to provision belief voluntarily must substitute repression R for legitimacy L—exactly the coercion-substitute row of the bilateral taxonomy, now derived from the public-good structure rather than asserted. Surveillance is what an order buys when it can no longer sell belief at scale.*

10.2 Markets in the Unverifiable

Proposition 10.3 (Schism as entry)**.** *Let m interpreters of an unverifiable counterparty D compete for the same legitimacy stock. Entry costs are near zero—the claims are unauditable—so the religious field fragments as participation grows: each increase in n raises the number of interpreters the shared stock can support, and denominations are the equilibrium granularity of a market in unverifiable claims.*

Proof. Each entrant earns a private share of the common stock while bearing only its entry cost; with entry costs near zero the field expands until each interpreter's share falls to that cost, so the equilibrium number of interpreters grows with the pool of participants n. □

Corollary 10.4 (Printing and reformation)**.** *A technology that lowers the cost of producing rival interpretations—the printing press above all—is an entry shock to the legitimacy market. It does not change the balance of force; it changes γ and the entry cost, and the incumbent order faces a legitimacy race it can lose without a single battlefield defeat.*

Remark 10.5. The signal monopoly of the bilateral theory survives only while the interpreter is singular. Multilaterally it is contestable, and contestability is exactly what the auditability ordering predicts: the less auditable the claim, the cheaper the entry, the finer the fragmentation.

11 Justice at Scale: Reform and Revolution

11.1 The Rebalancing Rule, Multilateral

The bilateral planner equated weighted marginal utilities across two actors. The multilateral planner equates them across n:

$$\lambda_i \frac{\partial u_i}{\partial s_i} \;=\; \lambda_j \frac{\partial u_j}{\partial s_j} \qquad \text{for all } i, j, \tag{20}$$

subject to incentive and participation constraints and to the requirement that the outcome be *coalition-proof*: no subset may prefer to defect from the settlement.

Definition 11.1 (Reform)**.** A *reform* is an adjustment of the ledger that remains within the core. A *revolution* is a jump to an allocation that the pre-existing core did not contain—typically because the core was empty.

Theorem 11.2 (Reform versus revolution)**.** *If the core is nonempty, incremental rebalancing along* (20) *suffices: under supermodularity the core is convex and hence path-connected, so any two core allocations are joined by a coalition-proof sequence. If the core is empty, no sequence of coalition-proof adjustments reaches a stable allocation, and the only stable endpoints are enforced by a center that did not previously exist, or are reached by a discontinuous jump—revolution, secession, or jubilee.*

Proof. For supermodular v the game is convex, its core is convex, and a coalition-proof path connects any two core allocations along (20). If the core is empty, Definition 5.1 and Proposition 5.5 give that no allocation is immune to blocking, so no coalition-proof step exists and the game itself must change—through a new enforcer or a discontinuous jump. □

Corollary 11.3 (Standing as reform-enabling)**.** *Extending standing to the full affected set enlarges N in the characteristic function, internalizing externalities, and can render a previously empty core nonempty. Representation is therefore not only just; it is the institutional precondition for settling disputes by reform rather than by force.*

11.2 The Adjustment Correspondence

The bilateral justice mechanism J carries over with the multilateral gap $G = \sum_i G_i$ and the Shapley target φ. Its effectiveness condition is unchanged in form: J must reduce G faster than the disequilibrium forces increase it. What is new is that J itself must be coalition-proof: a redistribution that some coalition can block is not a settlement but a truce.

12 Alliances, Spoils, and War

12.1 Alliance Parasitism

An alliance is a coalition with an external enemy. The external threat changes each member's exit calculation.

Theorem 12.1 (The common-enemy subsidy)**.** *Let members of an alliance face an external threat costing T_i per period when outside it, and let the hegemon extract τ_i from member i. Where the exit constraint binds, the sustainable extraction is*

$$\tau_i \;\leq\; 1 - \sqrt{2\big(\bar{a}_i - k_i - T_i\big)_+}, \tag{21}$$

which is strictly increasing in T_i: the threat lowers the value of the outside option, so members tolerate more extraction before leaving. A larger common enemy raises the hegemon's sustainable extraction from every ally.

Proof. Membership is preferred to exit iff the member's net gain under extraction exceeds the outside option net of exit cost, and the threat reduces the value of the outside option by T_i, lowering the continuation threshold exactly as an additional exit cost would. □

Corollary 12.2 (Defensive alliances become empires)**.** *A league formed against a common threat contains the mechanism of its own consolidation: while the threat persists, the strongest member can raise extraction from the others without any change in formal equality. The Delian League's transformation into the Athenian empire is* (21) *with T large and τ rising to its new ceiling.*

Corollary 12.3 (Dissolution after victory)**.** *When the threat ends, $T \to 0$ and the extraction ceiling collapses. The hegemon must then either democratize the alliance—raising members' voice to offset the fallen ceiling—or watch it dissolve.*

12.2 War

Remark 12.4 (War as process failure). War is the breakdown of the multilateral ledger's adjustment mechanism: the failure to reason together and to negotiate on equal terms as equal parties. Bilateral disputes are settleable because the affected set has two members who can, in principle, price the settlement. Multilateral disputes with coalitional uncertainty—who will side with whom, what each will demand after victory—make the bargaining set non-contractible: no side can verify that a rival will honor a concession once the balance shifts. The ledger equivalent is an empty core among the warring coalitions (proposition 5.5); force then determines what bargaining could not.

Proposition 12.5 (Spoils and cohesion)**.** *A hegemonic coalition survives only while each ally's post-extraction payoff exceeds its threat value outside the coalition. The hegemon's freedom of extraction is therefore bounded by the cohesion constraint, and the spoils of external predation must be distributed to hold the internal ledger together. Predatory coalitions are stable exactly to the degree that they redistribute.*

Corollary 12.6 (War raises extraction everywhere)**.** *War simultaneously (i) raises T for every member, raising ceilings (theorem 12.1); (ii) multiplies predators on every shared host, as warlords and contractors enter the race (theorem 8.1); and (iii) raises the center's hazard h, shortening its horizon (proposition 9.6). Wartime orders are therefore maximally extractive on every ledger at once, which is why they end, when they end, in resets—jubilees, revolutions, constitutions.*

13 Dynamics on Networks: Cascade and Systemic Collapse

13.1 The Coupled System

For each actor i, the multilateral dynamics extend the bilateral system by coupling through the graph:

$$\dot{K}_i = g_i K_i\Big(1 - \tfrac{K_i}{K_i^{\max}}\Big) + \sum_{e \ni i} n_i^e - \zeta_i K_i - \phi_i C_i - \sigma_K \sum_{j \sim i} \big(K_j - K_i\big)_+, \tag{22}$$

$$\dot{L}_i = \alpha(F_i - F^*) - \beta R_i + \gamma I_i + \sigma_L \sum_{j \sim i} \big(L_j - L_i\big)_+, \tag{23}$$

$$\dot{C}_i = \sigma\Big(D_i + P_i - \lambda L_i - \mu X_i + \sigma_C \sum_{j \sim i} C_j\Big), \tag{24}$$

$$\dot{s}_i = \kappa\big(\varphi_i - s_i\big), \tag{25}$$

where $j \sim i$ denotes adjacency in $\mathcal{G}$ and $\sigma_K, \sigma_L, \sigma_C \geq 0$ are contagion strengths. Capacity leaks from failing neighbors; legitimacy diffuses; conflict spreads.

13.2 Cascade

Linearize (22) about a state in which a fraction of nodes have failed. In the branching approximation on a configuration-model network with degree distribution $\{p_d\}$, define the excess degree

$$\kappa_d \;=\; \frac{\langle d^2 \rangle}{\langle d \rangle} - 1, \tag{26}$$

and let $\mu \in [0,1]$ be the mean probability that the loss of one neighbor pushes a node below its reproduction threshold.

Theorem 13.1 (Cascade)**.** *A failure cascade propagates systemically if and only if*

$$\kappa_d \, \sigma_K \, \mu \;>\; 1. \tag{27}$$

Below the threshold, cascades are contained; above it, a vanishing initial shock can consume the giant component.

Proof. A failed node transmits failure to each neighbor with probability $p = \sigma_K \mu$ in the linearized regime. Each failed node exposes on average κ_d further neighbors, so the expected new failures per failure are $p\,\kappa_d$; the branching process survives iff this exceeds one. □

Proposition 13.2 (Centrality amplifies predation)**.** *The cascade seeded by predation on a node of degree d has expected size proportional to $d\,\sigma_K \mu / (1 - \kappa_d \sigma_K \mu)$ in the subcritical regime. Predation on a hub is therefore systemically dangerous in proportion to its degree, while identical predation on a leaf is contained.*

Proof. The seed exposes d neighbors rather than the mean; the subsequent branching is the same subcritical process. □

Corollary 13.3 (Justice targets hubs)**.** *Justice mechanisms are systemically efficient when directed at highly central actors: protecting a hub's reproduction protects every relation it touches. Conversely, the most dangerous extractive strategy in a networked order is not broad extraction but* targeted *extraction—concentrating predation on high-centrality hosts—because it maximizes cascade risk per unit of extraction.*

Corollary 13.4 (The two faces of connectivity)**.** *Connectivity raises $\langle d \rangle$, which raises surplus through more relations, but also raises κ_d, which raises cascade risk. Orders therefore face a genuine trade-off absent from the bilateral theory: the integration that makes an empire productive is the same integration that makes its collapse systemic rather than local. The bilateral collapse was a death; the multilateral collapse is a contagion.*

Remark 13.5 (Caveat). Theorem 13.1 is a linearized branching result: it assumes locally tree-like structure, independent failures, and weak correlations. Clustering, assortativity, and intentional attack modify the threshold, not the form of the trade-off in Corollary 13.4.

13.3 Collapse and Renewal

Proposition 13.6 (Post-collapse sparsity)**.** *After a cascade, the rebuilt graph is sparser: trust decay raises the coordination friction γ of* (13) *and the organizational cost of forming relations. The post-collapse order is therefore smaller, more local, and slower to centralize—the formal content of the dark age that follows a systemic collapse, and of the localism of the orders that emerge from it.*

Proof. Cascade removes nodes and edges; rebuilding requires coalition formation under elevated coordination costs; Theorem 4.7 then bounds the reconstituted size by the reduced κ and elevated $\bar{a}$ of a low-trust environment. □

14 Scope and Objections

14.1 Complexity

The bilateral theory was tractable because two-party games have closed-form equilibria. The multilateral generalizations in this paper are exact where symmetry or convexity permits—the contribution equilibrium, the race, the coordination ceiling—and approximate where they do not—the cascade threshold, the public-good provision levels. The approximations are stated as such. The alternative, abandoning structure for numerical generality, trades a different kind of honesty for precision no historical data can support.

14.2 The Functionalism Objection

Does the center arise *because* it solves problems? The theory claims only that centers that solve the three problems sustain higher fees and persist; those that do not, do not. Selection among centers is historical; the constraint set is theoretical.

14.3 The Nationalism Objection

Nations are not in the primitives. They are coalitions whose stability is exactly what Section 5 and Section 7 are about: a nation is a coalition that has solved its coordination problem—lowered γ—and can therefore act with voice. The theory predicts that such coalitions are powerful precisely because coordination, not size, is the binding constraint.

14.4 Empirical Scope

The theory is comparative and structural. It yields testable implications: extraction falls with organizational density and rises with coordination cost; hegemonic extraction rises with the size of the common enemy; centers arise where predatory fragmentation is greatest; systemic collapses require networks above (27); revolutions follow shocks to γ rather than to shares. These are checkable against the historical record, and against the contemporary ledger of the companion essays.

15 Conclusion

The bilateral theory said: civilization is a ledger of reciprocal relations, and every durable order must keep it roughly balanced. This paper said: a civilization is not a pair of ledgers but a *network* of them, and the network has properties no pair possesses. Voluntary cooperation has a size limit set by technology—$m^{-1/D}$—and beyond that limit production is organized by command. Extraction has a ceiling set not only by each victim's ability to leave but by the victims' ability to act together, and that ability falls with the very size that makes them victims. Sharing a host multiplies the predators' effective impatience until patient raiders annihilate what a single patient raider would have conserved; the racers themselves prefer the monopoly they cannot arrange without a center, and so the center arises—a cartel among predators that also saves the host, charging a fee bounded by the value of both services, looting or building according to its own horizon. Belief that orders cannot provision at scale they must purchase with surveillance. Standing that does not match the footprint of effects produces injustice that no recorded ledger contains. And when the network's centrality is high enough, the death of a node becomes the death of its neighbors, and collapse stops being a local event and becomes a contagion. Four questions follow the bilateral four. The bilateral theory asked of every order: who gives, who gains, who cannot leave, who cannot speak. The multilateral theory adds: *who is missing from the ledger entirely*? The affected but unrepresented; the future generations moving silently in the atmosphere; the caregivers absent from the accounts; the peripheral nodes whose thresholds the capital's ledger never records. A civilization's depth is measured not by the size of its monuments nor the number of its citizens, but by how closely its standing follows its effects, how cheaply its coordination is made, how honestly its center is held to its own horizon, and whether its network is built so that no single failure need be everyone's.

A Summary of Formal Results

Table 1: Summary of the multilateral results and their bilateral antecedents.

Result	**Statement**	**Reading**
Command threshold	$\sum_i \alpha_i \geq 1 \Rightarrow$ no interior equilibrium	Hierarchy is a technology of scale
Homogeneity	c^* homogeneous of degree $1/D$	Surplus amplifies at scale
Overstretch	$u^* = \frac{1-\alpha}{\alpha}(\alpha\eta/m)^{1/D}$; finite m^*	Technology bounds political scale
Core and Shapley	v supermodular $\Rightarrow \varphi \in$ core	Fair share is coalition-proof average contribution
Standing	$A(e) \subseteq e$ required for legitimacy	Externalities are unledgered injustice
Coalitional patience	$\delta^*(n)$ nondecreasing in n	Scale demands monitoring
Coordination ceiling	$V(m) = mv/(1+\gamma(m-1))$; extraction rises in m when $\gamma \geq 1$	Large unorganized classes pay most
Predation race	$K^* = (1-mr)/(m+1)$; depletion iff $mr \geq 1$	Competition multiplies impatience
Predators' dilemma	$u^*(m)$ decreasing in m	States are predatory cartels
Center's fee	$f \leq$ conservation + cooperation + monitoring value	The state's services bound its take
Guardian	$r_C = r + h$; looting iff $r_C \geq 1$	Horizons, not virtues, build works
Legitimacy public good	I under-provided in n; R substitutes	Surveillance is belief at scale's failure
Alliance subsidy	$\partial\tau_i/\partial T_i > 0$	Enemies subsidize hegemons
Reform vs. revolution	Empty core $\Rightarrow$ no incremental settlement	Revolution as core emptiness
Cascade	Systemic iff $\kappa_d \sigma_K \mu > 1$	Collapse is a contagion above threshold

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